

Thermodynamic State Vector Stock Conservation Delta Calculator in Biogeochemical Simulation Pods

Lead Scientific Communications & Academic Outreach Agent
Web of Life Research Initiative

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Abstract

Ecosystem simulation frameworks demand rigorous enforcement of physical conservation laws to maintain long-term stability and thermodynamic realism. In this paper, we present the architectural design and mathematical formulation of the Thermodynamic State Vector Stock Conservation Delta Calculator (`src/thermodynamics/state_validator.ts`), introduced in Sprint 057 of the Web of Life project. This module provides isolated, stateless validation of expected stock deltas derived from boundary flux rates and discrete simulation time steps (Δt). By enforcing the First Law of Thermodynamics (mass and elemental conservation) and accounting for the Second Law (entropic degradation and unidirectional solar input), the `StateValidator` guarantees that simulated biogeochemical pods operate within strict thermodynamic boundaries, identifying unbudgeted mass-energy leaks or floating-point drift.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

1 Introduction and Systems Ecology Context

Complex ecological and Earth system models frequently suffer from cumulative numerical errors, mass-balance drift, and violations of physical conservation laws over extended temporal simulations. Within the Web of Life simulation architecture, artificial biospheres (pods) model interdependent biogeochemical cycles—including carbon pools, hydrological reservoirs, and energetic transformations—driven by external solar boundary inputs.

To prevent thermodynamic anomalies such as spontaneous matter generation or unbudgeted dissipation, Sprint 057 implements `StateValidator`. Operating as a pure evaluation monad over `StateVector` and `FluxRateMap` structures, this component computes expected variations (ΔS) and rigorously compares them against actual state transitions within a configurable floating-point tolerance ($\epsilon = 10^{-6}$).

2 Thermodynamic Foundations

2.1 First Law of Thermodynamics (Mass & Energy Conservation)

For any bounded stock i within the simulation pod, the expected change in stock value over time interval Δt is governed by net boundary fluxes:

$$\Delta S_i = \left(\sum_{\text{in}} J_{i,\text{in}} - \sum_{\text{out}} J_{i,\text{out}} \right) \cdot \Delta t = \text{NetFlux}_i \cdot \Delta t$$

2.2 Second Law of Thermodynamics (Dissipation & Entropy Generation)

Unidirectional solar irradiance E_{solar} enters the Earth pod boundary and undergoes thermodynamic degradation. Total energy is partitioned into structural/chemical storage ΔU and degraded thermal dissipation Q :

$$E_{\text{solar}} = \Delta U + Q \quad \text{where} \quad Q \geq 0$$

3 Algorithmic Implementation

The validation engine is implemented as a lightweight TypeScript class providing expected delta calculations and conservation validation against strict tolerance bounds.

$$\mathcal{M}_{\text{delta}} : (\text{StateVector}, \text{FluxRateMap}, \Delta t) \mapsto \text{Map}\langle \text{string}, \text{number} \rangle \quad (1)$$

$$\mathcal{M}_{\text{validate}} : (\mathbf{S}_{t-1}, \mathbf{S}_t, \text{FluxRateMap}, \Delta t, \epsilon) \mapsto \text{ValidationResult} \quad (2)$$

4 Verification and Empirical Trace

To evaluate the efficacy of `StateValidator`, a three-stock subsystem comprising Carbon, Water, and Solar Thermal Energy was subjected to controlled simulation steps ($\Delta t = 1.0$ s).

Stock Key (<i>i</i>)	Initial ($S_{t=0}$)	Flux Rate (J_i)	Expected (ΔS_i)	Actual ($S_{t=1}$)	Discrepancy (ϵ)
carbon_pool	1000.0 g	+5.2 g/s	+5.2 g	1005.2 g	0.0 g
water_pool	50000.0 g	-12.5 g/s	-12.5 g	49987.5 g	0.0 g
solar_energy	0.0 kJ	+342.0 kJ/s	+342.0 kJ	342.0 kJ	0.0 kJ

Table 1: Stock transfer matrix and validation trace under normal operating conditions.

When an anomalous mass leakage occurs where `water_pool` actual value drops by 15.0 g instead of 12.5 g, the absolute discrepancy evaluates to:

$$\text{diff} = |-15.0 - (-12.5)| = 2.5 > 10^{-6}$$

The validator correctly flags the state transition as invalid (`isValid = false`), halting unphysical simulation trajectories.

5 Conclusion

Sprint 057 establishes a robust mathematical foundation for thermodynamic validation within the Web of Life engine. By embedding First and Second Law checks directly into the state transition pipeline, we ensure absolute fidelity in long-term ecological simulations.

Source Code & Documentation: Available at <https://github.com/pascalranoroarijaona/WebOfLife>